Shashank Satish Adsule

Computer Science Engineer

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Education

- **Indian Institute of Techology Madras** — *Chennai*
  M.Tech Data Science and Artificial intelligence — *[2025-present]*
  Courses: machine Learning, NLP,
- **Indian Institute of Information Technology Pune** — *Pune*
  B.Tech – Computer science & Engineering [7.07 CGPA] — *[2020-2024]*
  Courses: *Operating system, OOPS, DSA, Machine Learning, Image processing*
- **Saint Mary's Jr college** — *Navi Mumbai*
  HSC – Bi-Focal (Electronics) [77.08 %] — *[2018-2020]*
  Courses: *Digital Logic, Basic Robotics,*
- **Christ Academy ICSE School** — *Navi Mumbai*
  SSC – Science [77.67 %] — *[2010-2018]*

Experience

- **Research Associate Intern** — *[Dec 2023-jul 2024]*
  Mantra Softech India Pvt Ltd (Bengaluru)

- Worked on Computer Vision models and Neural Networks for edge devices.
- Performed analysis and inference on various DNN models as part of research and development.

Projects worked on: YOLOv9, YOLOv5, MTCNN, FDDB for Face detection, Kneron (hardware) model integration.
Skills used: Python, C++, CMake, OpenCV, Pytorch, TensorFlow, Onnx, Linux, shell scripting, Bash, Computer vision, Image processing, Machine Learning, Docker

Projects

- **Stylized image using NST [Feb 2024]:** **[Github]**

- Aims for creating a stylized image from given original image and reference style image.
- Uses VGG19 CNN model for extracting features from both images.
- Style features are applied on content image through a neural network.

Technology used: VGG19, Pytorch, python, L-BFGS optimizer , OpenCV

- **CLI Morse code generator with audio generation [Jan 2024]:** **[Github]**

- Developed an .exe application written in C++ which can convert plain text to Morse code and vice versa
- This executable file supports both Windows and Linux operating systems.
- Also, it converts Morse code into a Morse audio file in .wav format.

- **Automatic number plate recognition using YOLO v8 and easyOCR [Nov 2023]:** **[Github]**

- Utilizes YOLO v8 for robust vehicle detection, efficiently locating license plates within images or video streams.
- Uses easyOCR for the extraction of relevant information from detected license plates.
- Stores the resulting information into a .csv file.

Technology used: python, YOLO, easyOCR, SORT, Roboflow, OpenCV

- **Image to sketch image using machine learning [Dec 2022]:** **[Github]**

- Utilizes K-Means Clustering algorithm for extracting desired colour quantization from original image. while retaining the original key features in the final sketch image using some image processing techniques.
- Aims to create a relevant and accurate sketch from the input image.

Technology used: python, K-means algorithm, OpenCV, Image processing, Streamlit.

Skill Summary

Operating system	: Linux, Windows
Programming Languages	: Bash, C, C++, CSS, HTML, Java, Java Script, Python, SQL,
Frameworks	: Flask, OpenCV, Onnx, Pytorch
Tools & Platforms	: Bitbucket, Docker, Git, GitHub, MySQL
Soft Skills	: Adaptability, Continuous Learning, Teamwork, Time Management

Awards & Certificates

- **Computer Language proficiency:**
  - C intermediate (Sololearn)
  - C++ intermediate (Sololearn)
  - Java (Hackerrank)
  - Scientific computing with python (freecodercamp)
  - SQL (Hackerrank)
- **Computer skills :**
  - problem solving (Hackerrank)
  - OpenCV Bootcamp (OpenCV University)
- **Other achievements**
  - Qualified National science Olympiad in dec-2016 & dec-2018 (with 1st rank in class in 2018)
  - Qualified International Mathematics Olympiad in jan-2017
  - Qualified International Science Olympiad in 2015 (with 3rd rank in class)